

Bash Script

Basic Shell Scripts

Output to screen

```
#!/bin/bash
# Simple output script
echo "Hello World"
```

[illegible]

Defining Tasks

```
#!/bin/bash
# Example of defining variables
a=Imran
b=Afzal
c='Linux class'
echo "My first name is $a"
echo "My surname is $b"
echo 'My surname is $c'
```

```

total 14140
-rw-r--r--. 1 root root 14443350 Jul  3 07:22 apache-tomcat-11.0.24.tar.gz
drwxr-xr-x. 9 root root 16384 Aug 26 10:46 apache-tomcat-11.0.24
-rw-r--r--. 1 root root 1 Aug 26 10:49 1
-rwxrwxrwx. 1 root root 56 Aug 27 09:59 bash.sh
-rwxrwxrwx. 1 root root 81 Aug 27 10:14 bash.ba
-rwxrwxrwx. 1 root root 83 Aug 27 10:19 smalltask.bash

[root@ip-172-31-38-241 ec2-user]# cat smalltask.bash
#!/bin/bash
# define small task
#
whoami
echo
pwd
echo
hostname
echo
ls -ltr
echo

[root@ip-172-31-38-241 ec2-user]# ll
total 14140
-rw-r--r--. 1 root root 1 Aug 26 10:49 1
drwxr-xr-x. 9 root root 16384 Aug 26 10:46 apache-tomcat-11.0.24
-rw-r--r--. 1 root root 14443350 Jul  3 07:22 apache-tomcat-11.0.24.tar.gz
-rwxrwxrwx. 1 root root 81 Aug 27 10:14 bash.ba
-rwxrwxrwx. 1 root root 56 Aug 27 09:59 bash.sh
-rwxrwxrwx. 1 root root 83 Aug 27 10:19 smalltask.bash

```

Read Input

Defining variables

```
#!/bin/bash
```

```
# Example of defining variables
```

```
a=vineeth
```

```
b=vemuri
```

```
c='Linux class'
```

```
echo "My first name is $a"
```

```
echo "My surname is $b"
```

```
echo 'My surname is $c'
```

Read Input

```
#!/bin/bash
```

```
# Read user input
```

```
echo "What is your first name?"
```

```
read a
```

```
echo
```

echo "What is your last name?"

read b

echo

echo Hello \$a \$b

```
[root@ip-172-31-38-241 ec2-user]# ls
1  apache-tomcat-11.0.24  apache-tomcat-11.0.24.tar.gz  bash.ba  bash.sh  name.bash  smalltask.bash
[root@ip-172-31-38-241 ec2-user]# vi name.bash
[root@ip-172-31-38-241 ec2-user]# ./name.bash
bash: ./name.bash: cannot execute: required file not found
[root@ip-172-31-38-241 ec2-user]# vi name.bash
[root@ip-172-31-38-241 ec2-user]# ./name.bash
what is your name?
sunith

what is you last name?
vineeth

Hello sunith vineeth
[root@ip-172-31-38-241 ec2-user]# cat name.bash
#!/bin/bash
# Read user input
echo "what is your name?"
read a
echo

echo "what is you last name?"
read b
echo

echo Hello $a $b
[root@ip-172-31-38-241 ec2-user]# |
```

Scripts to run commands within

```
#!/bin/bash
```

```
# Script to run commands within
```

```
clear
```

```
echo "Hello `whoami`"
```

```
echo
```

```
echo "Today is `date`"
```

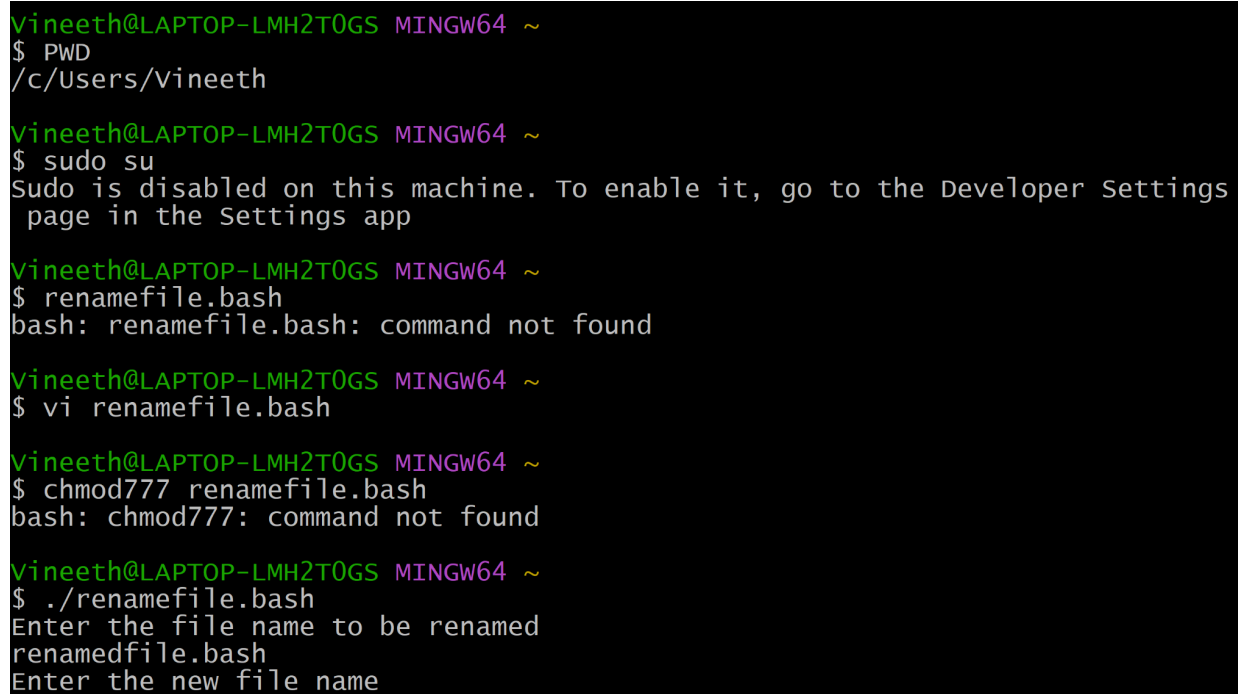
```
echo
```

```
echo "Number of user login: `who | wc -l`"
```

```
Echo
```

Read input and perform a task

```
#!/bin/bash
# This script will rename a file
echo Enter the file name to be renamed
read oldfilename
echo Enter the new file name
read newfilename
mv $oldfilename $newfilename
echo The file has been renamed as $newfilename
```



```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~
$ PWD
/c/Users/Vineeth

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~
$ sudo su
Sudo is disabled on this machine. To enable it, go to the Developer Settings
page in the Settings app

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~
$ renamefile.bash
bash: renamefile.bash: command not found

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~
$ vi renamefile.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~
$ chmod777 renamefile.bash
bash: chmod777: command not found

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~
$ ./renamefile.bash
Enter the file name to be renamed
renamedfile.bash
Enter the new file name
```

do-while Script

Script to run for a number of times

```
#!/bin/bash
c=1
while [ $c -le 5 ]
do
echo "Welcone $c times"
(( c++ ))
Done
```

```

ineeth@LAPTOP-LMH2T0GS MINGW64 ~
cd downloads

ineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
ssh -i "instance 1.pem" ec2-user@ec2-16-171-7-173.eu-north-1.compute.amazonaws.com
sh: connect to host ec2-16-171-7-173.eu-north-1.compute.amazonaws.com port 22: Connection timed out

ineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
ssh -i "instance 1.pem" ec2-user@ec2-16-171-7-173.eu-north-1.compute.amazonaws.com

#_
~\##### Amazon Linux 2023
~~~\#####
~~~\#####
~~~\###|
~~~\#/
~~~V~'-'> https://aws.amazon.com/linux/amazon-linux-2023
~~~~
~~~.-.-
~~~/_m/'-_-/_

last login: Thu Aug 27 18:30:21 2026 from 157.35.67.104
ec2-user@ip-172-31-38-241 ~]$
ec2-user@ip-172-31-38-241 ~]$ sudo su
root@ip-172-31-38-241 ec2-user]# vi numerooftimes.bash
root@ip-172-31-38-241 ec2-user]# ./numerooftimes.bash
ash: ./numerooftimes.bash: Permission denied
root@ip-172-31-38-241 ec2-user]# chmod 777 numerooftimes.bash
root@ip-172-31-38-241 ec2-user]# ./numerooftimes.bash
ash: ./numerooftimes.bash: cannot execute: required file not found
root@ip-172-31-38-241 ec2-user]# ./numerooftimes.bash
ash: ./numerooftimes.bash: cannot execute: required file not found
root@ip-172-31-38-241 ec2-user]# cat
C
root@ip-172-31-38-241 ec2-user]# cat numerooftimes.bash
#!/bin/bash

!/bin/bash
=1
while [ $c -le 5 ]
do
echo "welcome $c times"
( c++ )
done
root@ip-172-31-38-241 ec2-user]# ll
total 14152
-rw-r--r--. 1 root root      1 Aug 26 10:49 1
-rwxr-xr-x. 9 root root    16384 Aug 26 10:46 apache-tomcat-11.0.24
-rw-r--r--. 1 root root   14443350 Jul  3 07:22 apache-tomcat-11.0.24.tar.gz
-rwxrwxrwx. 1 root root     81 Aug 27 10:14 bash.ba
-rwxrwxrwx. 1 root root     56 Aug 27 09:59 bash.sh
-rw-r--r--. 1 root root    171 Aug 27 12:37 date.bash
-rwxrwxrwx. 1 root root    108 Aug 27 12:59 name.bash
-rwxrwxrwx. 1 root root     90 Aug 27 18:43 numerooftimes.bash
-rwxrwxrwx. 1 root root     83 Aug 27 10:19 smalltask.bash
root@ip-172-31-38-241 ec2-user]# ./numerooftimes.bash
ash: ./numerooftimes.bash: cannot execute: required file not found
root@ip-172-31-38-241 ec2-user]# vi numerooftimes.bash
root@ip-172-31-38-241 ec2-user]# ./numerooftimes.bash
/numerooftimes.bash: line 8: syntax error: unexpected end of file
root@ip-172-31-38-241 ec2-user]# vi numerooftimes.bash
root@ip-172-31-38-241 ec2-user]# ./numerooftimes.bash
welcome 1 times
welcome 2 times
welcome 3 times
welcome 4 times
welcome 5 times
/numerooftimes.bash: line 10: /root: Is a directory
/numerooftimes.bash: line 11: /root: Is a directory
/numerooftimes.bash: line 12: /root: Is a directory
/numerooftimes.bash: line 13: /root: Is a directory
root@ip-172-31-38-241 ec2-user]# |
```

do-while Script

done

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~  
$ pwd  
/c/Users/Vineeth  
  
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~  
$ cd downloads  
  
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads  
$ ssh -i "instance 1.pem" ec2-user@16.171.7.173  
  
      #_ _  
    ~\  #####  
   ~~ \  #####  
   ~~  \###|  
   ~~   \#/ V~' '->  
       ~~~  
       ~~~  
       ~..-.  
        \ /  
       _/m/' -/  
  
Amazon Linux 2023  
  
https://aws.amazon.com/linux/amazon-linux-2023  
  
Last login: Fri Aug 28 06:33:58 2026 from 103.160.27.199  
[ec2-user@ip-172-31-38-241 ~]$ sudo su  
[root@ip-172-31-38-241 ec2-user]# v while.bash  
bash: v: command not found  
[root@ip-172-31-38-241 ec2-user]# vi while.bash  
[root@ip-172-31-38-241 ec2-user]# chmod 777 while.bash  
[root@ip-172-31-38-241 ec2-user]# ./while.bash  
welcome 1 times  
welcome 2 times  
welcome 3 times  
welcome 4 times  
welcome 5 times  
[root@ip-172-31-38-241 ec2-user]#
```

Script to run for a number of seconds

```
#!/bin/bash
count=0
num=10
while [ $count -lt 10 ]
do
echo
echo $num seconds left to stop this process $1
echo
sleep 1
num=`expr $num - 1`
count=`expr $count + 1`
done
echo
echo $1 process is stopped!!!
echo
```

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~
$ pwd
/c/Users/Vineeth

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~
$ cd downloads

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ ssh -i "instance 1.pem" ec2-user@16.171.7.173

#
~\##### Amazon Linux 2023
~\#####\
~\###|
~\#/\
~V~'-'> https://aws.amazon.com/linux/amazon-linux-2023
~
~
~\m/'
~

Last login: Fri Aug 28 06:33:58 2026 from 103.160.27.199
[ec2-user@ip-172-31-38-241 ~]$ sudo su
[root@ip-172-31-38-241 ec2-user]# v while.bash
bash: v: command not found
[root@ip-172-31-38-241 ec2-user]# vi while.bash
[root@ip-172-31-38-241 ec2-user]# chmod 777 while.bash
[root@ip-172-31-38-241 ec2-user]# ./while.bash
welcome 1 times
welcome 2 times
welcome 3 times
welcome 4 times
welcome 5 times
[root@ip-172-31-38-241 ec2-user]# client_loop: send disconnect: Connection reset by peer

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ vi whilerunsec.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ chmod 777 whilerunsec.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ chmod ./whilerunsec.bash
chmod: missing operand after './whilerunsec.bash'
Try 'chmod --help' for more information.

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ ./whilerunsec.bash

10 seconds left to stop this process

9 seconds left to stop this process

8 seconds left to stop this process

7 seconds left to stop this process

6 seconds left to stop this process

5 seconds left to stop this process

4 seconds left to stop this process

3 seconds left to stop this process

2 seconds left to stop this process

1 seconds left to stop this process

process is stopped!!!

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ |
```

For loop Scripts:

Simple for loop output

```
#!/bin/bash
for i in 1 2 3 4 5
do
echo "Welcome $i times"
done
```

A terminal window with a black background and yellow and green text. The prompt is 'Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads'. The user enters '\$ vi loop.bash', '\$ chmod 777 loop.bash', and '\$./loop.bash'. The output shows five lines: 'welcome 1 times', 'welcome 2 times', 'welcome 3 times', 'welcome 4 times', and 'welcome 5 times'. The prompt returns to '\$ |' at the bottom.

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ vi loop.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ chmod 777 loop.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ ./loop.bash
welcome 1 times
welcome 2 times
welcome 3 times
welcome 4 times
welcome 5 times

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ |
```

Simple for loop output

```
#!/bin/bash
for i in eat run jump play
do
echo See vineeth $i
done
```

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ vi foroutput.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ chmod 777 foroutput.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ ./foroutput.bash
see vineeth eat
see vineeth run
see vineeth jump
see vineeth play

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ |
```

For loop to create 5 files named 1-5

```
#!/bin/bash
for i in {1..5}
do
touch $i
Done
```

```

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ vi forfile.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ chmod 777 forfile.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ ./forfile.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ ls -ef
ls: unknown option -- e
Try 'ls --help' for more information.

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ ls -ltr
total 5378114
-rwxr-xr-x 1 Vineeth 197121 1343320 Mar 2 2022 ChromeSetup.exe*
-rwxr-xr-x 1 Vineeth 197121 1343320 Mar 2 2022 'ChromeSetup (1).exe'*
-rwxr-xr-x 1 Vineeth 197121 1414600 Jun 12 2022 'ChromeSetup (2).exe'*
-rw-r--r-- 1 Vineeth 197121 41371649 Apr 6 2024 'Vikas Koukuntla-Apnabiodata (1).pdf'
-rw-r--r-- 1 Vineeth 197121 282 Sep 25 2025 desktop.ini
-rwxr-xr-x 1 Vineeth 197121 10869176 Oct 7 2025 'ChromeSetup (3).exe'*
-rwxr-xr-x 1 Vineeth 197121 1104952 Oct 29 2025 'WhatsApp Installer.exe'*
-rw-r--r-- 1 Vineeth 197121 64758 Feb 18 2026 'VINEETH KAUKUNTALA (1).pdf'
-rw-r--r-- 1 Vineeth 197121 135138 Mar 19 20:07 'WhatsApp Image 2026-03-19 at 8.06.47 PM.jpeg'
-rw-r--r-- 1 Vineeth 197121 115982 Mar 19 20:10 'WhatsApp Image 2026-03-19 at 8.08.59 PM.jpeg'
-rw-r--r-- 1 Vineeth 197121 319534 Mar 19 20:25 'Bio Data Doc in Denim Blue Grey Simple and Minimal Style.pdf'
-rwxr-xr-x 1 Vineeth 197121 0 Mar 19 20:32 'Bio Data Doc in Denim Blue Grey Simple and Minimal Style/'
-rw-r--r-- 1 Vineeth 197121 586700 Apr 10 13:09 Marksmemo_A248C1188_3_DEC-2025.pdf
-rw-r--r-- 1 Vineeth 197121 162 Apr 29 09:05 '~$neeth_ATS_Optimised_Resume.docx'
-rw-r--r-- 1 Vineeth 197121 148054 Jul 9 22:23 FEE.pdf
-rwxr-xr-x 1 Vineeth 197121 49381480 Aug 10 11:27 'Git-2.37.1-64-bit.exe*'
-rw-r--r-- 1 Vineeth 197121 3853824 Aug 10 11:44 putty-64bit-0.84-installer.msi
-rw-r--r-- 1 Vineeth 197121 27785951 Aug 10 11:49 MobaXterm_Portable_v22.1.zip
-rwxr-xr-x 1 Vineeth 197121 6933184 Aug 10 12:00 npp.8.9.7.Installer.x64.exe*
-rwxr-xr-x 1 Vineeth 197121 237798672 Aug 10 12:04 VSCodeUserSetup-x64-1.132.0.exe*
-rwxr-xr-x 1 Vineeth 197121 28701168 Aug 10 12:27 BCompare-5.2.5.32528.exe*
-rwxr-xr-x 1 Vineeth 197121 287670872 Aug 10 12:56 VMware-Workstation-Full-26H1-25388281.exe*
-rw-r--r-- 1 Vineeth 197121 4762707968 Aug 10 13:29 ubuntu-22.04.5-desktop-amd64.iso
-rw-r--r-- 1 Vineeth 197121 15625 Aug 10 14:53 Self-Introduction-Experienced-Candidate.docx
-rw-r--r-- 1 Vineeth 197121 15625 Aug 11 10:05 'Self-Introduction-Experienced-Candidate (1).docx'
-rw-r--r-- 1 Vineeth 197121 0 Aug 12 15:42 vineeth
-rw-r--r-- 1 Vineeth 197121 92933 Aug 12 22:15 linux-and-itsm-and-patchmanagement-and-commands.docx
-rw-r--r-- 1 Vineeth 197121 6497115 Aug 13 12:58 'Linux Commands.pdf'
-rw-r--r-- 1 Vineeth 197121 8613373 Aug 13 13:40 'Linux Commands.docx'
-rw-r--r-- 1 Vineeth 197121 1974045 Aug 14 00:07 Gemini_Generated_Image_sh8bf8sh8bf8sh8b.png
-rw-r--r-- 1 Vineeth 197121 23384064 Aug 17 20:28 chromeremotedesktophost.msi
-rw-r--r-- 1 Vineeth 197121 1678 Aug 18 13:05 V.pem
-rw-r--r-- 1 Vineeth 197121 1678 Aug 21 01:24 test.pem
-rw-r--r-- 1 Vineeth 197121 1678 Aug 24 12:53 text2.pem
-rw-r--r-- 1 Vineeth 197121 2872681 Aug 25 15:31 'Web^0App_servers_tasks.docx'
-rw-r--r-- 1 Vineeth 197121 1674 Aug 25 18:23 'instance 1.pem'
-rwxr-xr-x 1 Vineeth 197121 209 Aug 28 14:56 whilerunsec.bash*
-rwxr-xr-x 1 Vineeth 197121 63 Aug 28 15:04 loop.bash*
-rwxr-xr-x 1 Vineeth 197121 71 Aug 28 15:19 foroutput.bash*
-rwxr-xr-x 1 Vineeth 197121 54 Aug 28 15:31 forfile.bash*
-rw-r--r-- 1 Vineeth 197121 0 Aug 28 15:31 1
-rw-r--r-- 1 Vineeth 197121 0 Aug 28 15:31 2
-rw-r--r-- 1 Vineeth 197121 0 Aug 28 15:31 3
-rw-r--r-- 1 Vineeth 197121 0 Aug 28 15:31 4
-rw-r--r-- 1 Vineeth 197121 0 Aug 28 15:31 5

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$

```

Specify days in for loop

```
#!/bin/bash
```

```
i=1
```

```
for day in Mon Tue Wed Thu Fri
```

```
do
```

```
echo "Weekday $((i++)) : $day"
```

```
Done
```

```
Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ vi fordayloop.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ chmod 777 fordayloop.bash

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ ./fordayloop.bash
./fordayloop.bash: line 1: i#!/bin/bash: No such file or directory
Weekday 1 : Mon
Weekday 2 : Tue
Weekday 3 : Wed
Weekday 4 : Thu
Weekday 5 : Fri

Vineeth@LAPTOP-LMH2T0GS MINGW64 ~/downloads
$ |
```

List all users one by one from /etc/passwd file

```
#!/bin/bash
i=1
for username in `awk -F: '{print $1}' /etc/passwd`
do
echo "Username $((i++)) : $username"
Done
```

```
[root@ip-172-31-7-191 Scripts]#  
[root@ip-172-31-7-191 Scripts]# chmod +x forlistusers.bash  
[root@ip-172-31-7-191 Scripts]#  
[root@ip-172-31-7-191 Scripts]# ./forlistusers.bash  
Username 1 : root  
Username 2 : bin  
Username 3 : daemon  
Username 4 : adm  
Username 5 : lp  
Username 6 : sync  
Username 7 : shutdown  
Username 8 : halt  
Username 9 : mail  
Username 10 : operator  
Username 11 : games  
Username 12 : ftp  
Username 13 : nobody  
Username 14 : dbus  
Username 15 : systemd-network  
Username 16 : systemd-oom  
Username 17 : systemd-resolve  
Username 18 : rpc  
Username 19 : libstoragemgmt  
Username 20 : systemd-coredump  
Username 21 : systemd-timesync  
Username 22 : sshd  
Username 23 : chrony
```